

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0653

Harrogate 138 kV
July 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0653. The Interconnection Customer proposes a 327.4 MW natural gas combined cycle facility interconnecting at the Harrogate 138 kV substation in the Virginia area. The request was received on July 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 218 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,168,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chancellor Row Substation-Natchez 138 kV	Skaggs Island Jct 2-Mitchell Dam 345 kV	121.6	213	18.4
Westcoast Cellufibre-Chipola 138 kV	Mitchell Dam-Skaggs Island Jct 2 345 kV	120.9	257	17.5
Highland City-Centralia 138 kV line	Skaggs Island Jct 2-Mitchell Dam 345 kV L	106.4	245	17.0
Westrock Sutheast Llc-Pacifica 138 kV line	SkaggsIsland Jct 2-Mitchell Dam 345 kV	102.1	166	17.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chancellor Row Substation-Natchez 138 kV	Rebuild line with higher-capacity conductor	\$2,278,000	14% (\$321,000)
Westcoast Cellufibre-Chipola 138 kV	Rebuild line with higher-capacity conductor	\$2,077,000	10% (\$207,000)
Highland City-Centralia 138 kV line	Reconductor line	\$1,004,000	78% (\$786,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Westrock Southeast Llc-Pacifica 138 kV	Reconductor line	\$1,091,000	78% (\$854,000)
	Total allocated to project		\$2,168,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.